

ABHISHEK SHARMA

MS by Research Indian Institute of Technology (IIT) Dharwad

Computer Vision · Time Series Modeling · Large Language Models (LLMs) · Multimodal Learning

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Technical Skills

- **Programming Languages** : Python (*primary*), C++, C, Java, HTML, CSS
- **ML / DL Frameworks** : PyTorch, TensorFlow, Keras, Hugging Face Transformers, LangChain, LlamaIndex
- **Gen AI & LLM Stack** : Prompt Engineering, RAG Pipelines, Fine-Tuning, RLHF, OpenAI API, Anthropic API
- **ML Libraries & Tools** : NumPy, Pandas, Scikit-learn, OpenCV, Matplotlib, SciPy, AWS Deployment
- **Robotics & Simulation** : ROS, RViz, Gazebo
- **MLOps & Infrastructure** : Docker, Git, Weights & Biases (W&B), CUDA, GPU Clusters (A100 / H100)
- **Development Environments** : VS Code, Jupyter Notebook, Google Colab
- **Operating Systems** : Ubuntu / Linux, Windows, macOS

Research Publications

- **Abhishek Sharma**, Sachin Maurya, Dr. Achyut Mani Tripathi, Dr. Kedar Khandeparkar.
“**MaHaWave-Net: A Lightweight Multi-Scale Model for Fine-Grained Medical Image Segmentation.**”
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2026).
Accepted CORE Rank: A* Conference | H-Index: 137
- **Abhishek Sharma**, Sachin Maurya, Dr. Achyut Mani Tripathi, Dr. Kedar Khandeparkar.
“**WaveletMamba: Harnessing Learnable Multi-Scale State Space Dynamics for Time Series Forecasting.**”
Neural Networks — Elsevier Journal 2026.
Accepted CORE Rank: Q1 Journal | Impact Factor: 7.3+

Education

Indian Institute of Technology (IIT) Dharwad

Aug 2024 – Present

MS by Research | Department of Computer Science and Engineering

Research Areas: Computer Vision · Time Series Modeling · Multimodal Learning · LLMs & VLMs

Funding: Teaching Assistantship (Institute Recommended) | GATE 2024: 95.64 percentile

CGPA: 7.87 / 10

G.V. Acharya Institute of Engineering and Technology

Jul 2018 – Jun 2022

Bachelor of Engineering | Department of Computer Science and Engineering

Affiliated to University of Mumbai

CGPA: 8.07 / 10

Work Experience

boAt Lifestyle R&D

Bengaluru, Karnataka

Sept 2022 – July 2023

Consultant Embedded Software Engineer

- Engineered real-time embedded software in **C/C++** for hearable IoT devices (TWS earbuds, neckbands), optimizing latency-critical signal pipelines directly applicable to **on-device AI/ML inference**.
- Built **Python-based automation and testing frameworks**, reducing manual QA overhead and accelerating release cycles — skills transferable to **ML experiment automation and CI/CD for model pipelines**.
- Maintained **BLE and Wi-Fi connectivity stacks**, gaining hands-on expertise in low-power protocols critical for **edge AI deployment** scenarios.
- Performed **system-level debugging** with protocol analyzers across hardware, firmware, and QA teams — developing root-cause analysis skills directly applicable to **ML pipeline and model debugging**.

- Developed **robotic control and automation software** using **Python and ROS**, building a strong foundation in **perception-action pipelines** foundational to embodied AI and autonomous systems.
- Leveraged **ROS tools** (RViz, rqt, rosbag) for robot navigation, sensor visualization, and data recording — skills directly transferable to **multimodal dataset curation and annotation pipelines**.
- Integrated hardware-software modules with **sensor fusion** techniques, gaining real-time data processing experience relevant to **computer vision and time-series modeling research**.
- Applied structured testing and validation methodology across full development cycles, mirroring rigorous **ML model evaluation and benchmarking workflows**.

Research Projects

Enhanced Spatial Feature Selective Mamba Framework for Image Segmentation Jan 2025 – May 2025
Research Project | IIT Dharwad *Python · PyTorch · CUDA · State Space Models (Mamba)*

- Designed **EFSVM-UNet**, a novel **State Space Model (Mamba)** architecture with a lightweight **Spatial Feature Selective (SFS)** module, advancing medical image segmentation beyond CNN and Transformer baselines.
- Proposed a custom **feature importance mask** and **task-specific loss function** for adaptive suppression of uninformative spatial features, improving generalization and reducing overfitting.
- Benchmarked against **SOTA models** across **6 public datasets** (ISIC 2017/18, Kvasir-SEG, CVC-ClinicDB/ColonDB, ETIS), achieving superior **Dice, IoU, and boundary accuracy** metrics.
- Conducted rigorous **ablation studies** validating each architectural component — research methodology aligned with **peer-reviewed publication standards (ICASSP A*)**.

Vision-Language Hybrid Framework for Automated Radiology Report Generation Jul 2025 – Sep 2025
Research Project | IIT Dharwad *Python · PyTorch · Transformers · VLMs · CUDA*

- Architected a **multimodal Vision-Language Model (VLM)** pipeline combining **dual visual encoders (ResNet101 + AdaMamba)** and an **MLP-Mixer** for global-local feature fusion, fed into a **Transformer decoder with expert routing** for clinical report synthesis.
- Proposed a **parameter-efficient lightweight decoder** to reduce Transformer FLOPs and inference latency — directly applicable to **production LLM/VLM deployment and optimization**.
- Demonstrated **cross-modal transfer learning** by reusing pretrained **Mamba** visual backbones across segmentation and report generation tasks, showcasing architectural reusability.
- Outperformed **R2Gen baselines** on **IU X-ray: BLEU-1 0.505, ROUGE-L 0.383, METEOR 0.190** — evaluated using **NLG metrics** consistent with LLM benchmarking frameworks.

Achievements & Activities

- **Certification — Generative AI & Agentic AI** — Completed as part of the **AI-Native Software Engineering Program**, Indian Institute of Technology (IIT) Kharagpur. Covered LLM architectures, RAG pipelines, autonomous agent design, and production-grade Gen AI system development.
- **Foundational Maths for ML: Statistics** — Certification from **C-MInDS, IIT Bombay**. Covered advanced probability, concentration inequalities, LLN, CLT, parameter estimation (MLE, MAP), linear regression, bias-variance tradeoff, and regularization techniques (Lasso & Ridge).
- **IEEE ICEI 2024 / 2025 Volunteer** — Served at the 2nd and 3rd IEEE Conference on Engineering Informatics at IIT Dharwad, contributing to a top-tier academic research forum.

References / Recommendation

- **Dr. Achyut Mani Tripathi** — Assistant Professor, IIT Dharwad
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- **Dr. Kedar Vithal Khandeparkar** — Assistant Professor, IIT Dharwad
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